TRAINER KEY. DO NOT DISTRIBUTE. Day 2, Session 1: What would have happened anyway? (difference-in-differences)

Source: H4-A (page 1). Key figures: participants 1,432 to 763 (-669); comparison 2,258 to 2,401 (+143); DiD = -669 - 143 = **-812 AED**, 95% interval -833 to -792; the 1,000 AED rule is not met by the estimate or the interval. Parallel trends is an assumption: the data have one pre-period and cannot show earlier trends.

Underlined = claim the source supports. ~~Struck out~~ = claim the source does not support. Green boxes = model answers (accept equivalent wording).

Page 1 | H4-A: The city difference-in-differences (DiD) study summary

GreenWaste is a fictional training case. Use this study summary as your source for today's questions.

§1 | Who, what and when?

GreenWaste provides equipment subsidies, installation help and staff training. The city study follows 10,000 businesses: 4,794 participant businesses and 5,206 comparison businesses that did not receive GreenWaste. Eligibility followed an efficiency-score rule, not a lottery. Annual waste management cost per business, in AED, was recorded once before GreenWaste and once 12 months after. These are city records. They are not the separate pilot of 400 businesses.

§2 | What was compared?

The participant businesses' mean cost went from 1,432 to 763 AED, a change of -669 AED. The comparison businesses' mean cost went from 2,258 to 2,401 AED, a change of +143 AED. The difference-in-differences (DiD) estimate is the participants' change minus the comparison businesses' change, which is about **-812 AED**. The calculations use the exact records, and the figures shown are rounded. This is the difference between the two changes. It is not the participant businesses' own change, and it is not the gap between the two groups after the programme.

Annual waste management cost in the city study: change in participant businesses compared with change in comparison businesses (AED)

Row	Coefficient (AED)	95% interval (AED)	p-value
Comparison businesses, before GreenWaste (intercept)	2,258	2,242 to 2,274	< 0.001
Change in comparison businesses (after)	143	129 to 157	< 0.001
Participant businesses minus comparison businesses, before GreenWaste (took part)	-826	-847 to -804	< 0.001
Difference between the two changes (after × took part)	-812	-833 to -792	< 0.001

The table uses 10,000 city businesses with two cost measurements each (before and 12 months after). The model is cost ~ after × took part. The 95% intervals allow for the two measurements on each business. The reference row is the comparison businesses before the programme. The last row is the difference between the two changes.

§3 | What is needed for a causal reading?

The parallel trends assumption means that, without GreenWaste, the two groups would have had the same mean change in cost over this period. They do not need to start at the same cost. Under this assumption, the participant businesses' mean cost after 12 months without GreenWaste would be about 1,575 AED, not the 763 AED we observe. A causal reading also needs comparable measurement, stable group membership and no important effect of the programme on the comparison businesses. A competing change that affected the groups differently could undermine the reading.

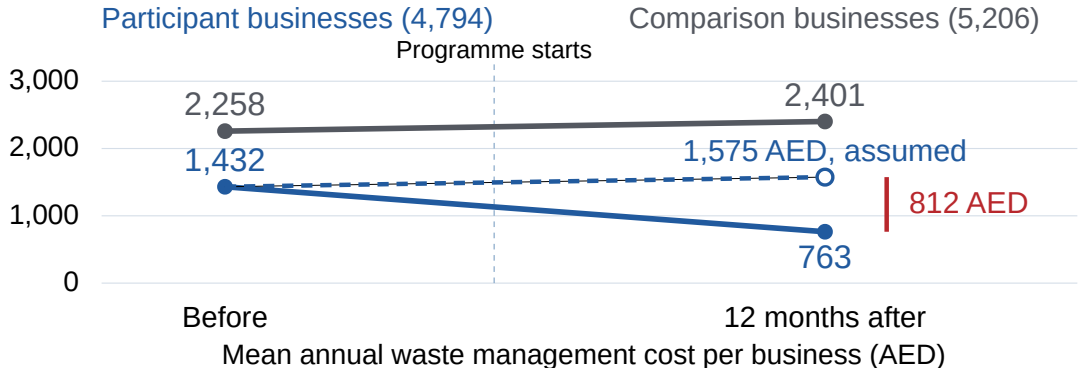
§4 | What can this study summary show?

One measurement before GreenWaste cannot show earlier trends. The histories A and B on page 2 are hypothetical. They are not additional observations. The 95% interval for the difference between the two changes is a reduction of 792 to 833 AED, if the model’s assumptions hold. A precise interval does not verify parallel trends. Both the estimate and the interval are smaller than our rule of at least a 1,000 AED reduction in annual waste management cost per business per year. This table alone does not show effects for the whole country or whether GreenWaste is good value for money, because the full programme costs are not given.

Source: GreenWaste city data (10,000 businesses, measured before and 12 months after). The regression uses the same businesses at both times and allows for the repeated measurements on each business. The paragraphs and table above are the material to inspect.

Page 2 | H4-A: Figures for reference

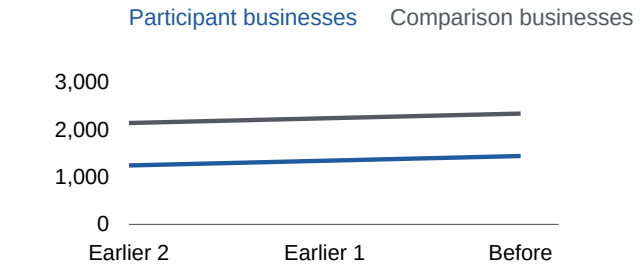
Observed means and an assumed path for participant businesses without GreenWaste



Solid lines join the observed means. The dashed line shows the path the participant businesses would have followed under the parallel trends assumption.

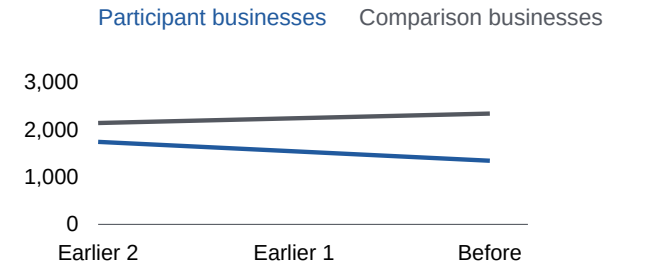
Two hypothetical earlier histories, A and B

A



ean annual waste management cost (AED), all before the programn

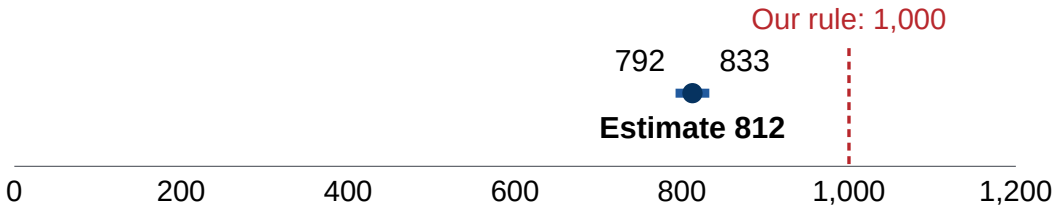
B



ean annual waste management cost (AED), all before the programn

Both histories use the same scales and both end before the programme. Groups can start at different levels and still change by similar amounts. Neither history comes from the GreenWaste data.

The difference between the two changes compared with our rule



The interval shows uncertainty about the mean difference between the two changes, if the model's assumptions hold. It does not show the savings of each business, and it does not allow for every possible source of bias.

Page 3 | Q1 and Q2: work on these as we teach each idea

Q1 | Four numbers, two changes, one difference between the changes

Use §2 on page 1.

a. Calculate the change (after minus before) for each group in the table.

Group	Before (AED)	After (AED)	Change (AED)
Participant businesses	1,432	763	**-669**
Comparison businesses	2,258	2,401	**+143**

b. Subtract the comparison businesses' change from the participant businesses' change.

Participant change minus comparison change (AED): **-812 (a reduction of 812 AED)**

c. Label the four observed means on the figure and connect each group's two points. After we explain the dashed line, add the assumed participant outcome and label it "assumed".



Q2 | Find the difference between the two changes in the regression table

a. Circle its row in the table on page 1.

b. Explain in your own words what its sign, its units, the comparison and the group of businesses tell you.

> **Sign:** negative = lower cost. **Units:** AED per business per year (mean annual waste management cost). **Comparison:** change in participants minus change in comparison businesses. **Group:** the 10,000 city businesses (4,794 participants, 5,206 comparison). Participants' mean cost fell 812 AED more than the comparison group's, if the model assumptions hold.

c. What does each of the other three rows show?

> Row 1: comparison businesses' mean cost before (2,258). Row 2: change in comparison businesses (+143; their cost rose). Row 3: participants minus comparison before GreenWaste (-826; participants started lower, so the groups were not similar at the start).

d. Why is the last row not the participant businesses' own change of -669 AED?

> The last row compares changes: participants' -669 minus comparison's +143. It removes the comparison group's +143 change as an estimate of what participants would have seen anyway, so it is larger than the participants' own change.

e. Box the 95% interval in the last row. Say what it describes.

> Box -833 to -792. With the model's assumptions, the mean difference between the two changes is plausibly a reduction of 792 to 833 AED. It describes uncertainty about the mean, not individual businesses and not bias.

f. Read the p-value in that row. Say what it does and does not show.

> p < 0.001: a difference between the changes this large would be very unlikely if the true mean difference were zero (under the model). It does not show that parallel trends holds, that the effect is causal, or that it meets the 1,000 AED rule.

Page 4 | Q3 and Q4: how credible is the result, and what should we advise?

Q3 | What would have happened without GreenWaste?

Use §3 and §4 on page 1 and the figures on page 2.

a. Explain the parallel trends assumption for GreenWaste in your own words. The groups do not need to start at the same cost.

> Without GreenWaste, participants' cost would have changed by the same amount as the comparison group's (+143), so participants would be about 1,575 AED instead of 763. Starting levels can differ; the changes must match.

b. Which hypothetical history, A or B, makes you hesitate? Describe what looks different.

> Accept either with a reason. History B (lines diverging before the programme) should make you hesitate; A (similar slopes) is reassuring. Look for lines that are not parallel before the start.

c. What does the more reassuring history still not prove?

> Similar earlier trends do not prove the groups would have continued in parallel after the start. A competing change at the same time could still affect them differently.

d. Could the GreenWaste data show earlier trends?

> No. There is one measurement before and one after, so no earlier trend can be seen. Histories A and B are hypothetical.

e. Request one specific piece of evidence about earlier costs or about a competing change, and say why it matters.

> Examples: monthly or annual costs for both groups for several periods before GreenWaste; or evidence of a change (price change, new regulation, local service change) that hit one group more than the other. Why: it tests parallel trends or rules out a competing explanation.

Q4 | Some reduction, or a large enough reduction?

Use the interval on page 2.

a. How do the estimate and the interval compare with our rule of at least a 1,000 AED reduction in annual waste management cost per business per year?

> The estimate (812) and the whole interval (792 to 833) are below the 1,000 AED rule. There is a reduction, but not one that meets the rule.

b. What does a narrow interval still leave unresolved?

> It does not resolve whether the causal assumption holds (parallel trends), measurement comparability, competing changes, or the programme costs. Precision is not validity.

Check the source, then give your own reading

In H4-A (the difference-in-differences study summary), circle the last row of the table and its interval, underline the assumption and mark the limit of the data. Compare your marks with a partner.

Keep H4-A open, and close the AI responses and your partner’s comments. Write one interpretation that H4-A supports and one specific evidence request. How could the answer change your advice?

> Example: "The data show participants' costs fell about 812 AED more than comparison costs, but the causal reading depends on parallel trends, which these data cannot test, and the estimate does not meet the 1,000 AED rule. I would ask for several pre-programme cost periods for both groups. If the earlier trends differ, my advice would be to delay a decision." Accept any supported interpretation plus a specific request.

Page 5 | AI exercise: check the numbers and the claims

Part A: Mark the response

Naive prompt (a prompt with no context, as a busy person might type it): “Explain the regression table to a policy adviser considering GreenWaste expansion. Say what they should check.” This response was written for the exercise. It is not a real AI answer. It mixes accurate readings with questionable conclusions. Do not run this prompt yourself.

> The regression reports a difference-in-differences estimate: participant businesses' annual costs fell by 669 AED while comparison businesses' costs rose by 143 AED. The difference between the two changes is a reduction of 812 AED per participant business, with a 95% interval of 792 to 833 AED and p < 0.001. Comparing changes is stronger than simply comparing cost levels after the programme.
>
> The different starting costs are accounted for, ~~and parallel trends is satisfied because both groups were followed over the same period.~~ ~~The small p-value and interval excluding zero confirm that the finding is robust and GreenWaste caused the saving.~~
>
> Implementation should still be monitored, particularly in new districts. ~~The result clears the city authority's 1,000 AED minimum, so expansion is justified;~~ ~~the precise estimate gives confidence that a wider rollout will meet this target.~~

Underline the claims that H4-A (the difference-in-differences study summary) supports. Cross out the claims that H4-A does not support. Check H4-A on page 1, not whether the writing sounds confident.

Part B: Run the improved prompt, then compare

Improved prompt

Explain H4-A to a policy adviser considering GreenWaste expansion. The outcome is annual waste management cost in AED per business. There are 4,794 participant businesses and 5,206 comparison businesses that did not receive GreenWaste. They were selected by the score rule, not by a lottery. The difference-in-differences estimate is a reduction of 812 AED, with a 95% interval of 792 to 833 AED. The data have one measurement before GreenWaste and one 12 months after, so they cannot show earlier trends. Parallel trends is an assumption, not a reported check. Our rule is a reduction of at least 1,000 AED per business per year.

Ask any essential clarifying questions first. Then give three short points: what the comparison and the last row of the table show; what must be credible for a causal reading; and whether the estimate and interval meet the rule. Cite H4-A and request specific missing evidence. Do not invent a test for earlier trends. Do not treat a narrow interval as proof that the result is robust or applies nationwide.

Copy this prompt and H4-A into Microsoft Copilot, and keep your AI good-practice card beside you. If you are not using Copilot, read the prepared response on page 6 after you finish Part A.

a. Which claim improved?

> Look for: it no longer says parallel trends is satisfied or proven; it separates the comparison, the causal assumption and the decision rule; it notes that 812 and the interval are below 1,000.

b. What still needs checking against H4-A?

> Check each number against §2 and the table (-669, +143, -812, 792 to 833); that it does not invent a test for earlier trends; that "robust" and national claims are not reintroduced; the 1,000 AED comparison.

Page 6 | Prepared response

Use this response if you are not using Copilot. Read it after you have marked the response on page 5. It was written for the exercise in reply to the improved prompt. It is not additional evidence from the study. An AI tool may answer differently.

> Before I advise on expansion, I would ask for earlier cost measurements in both groups and for information on other changes that might have affected them differently. I would also ask whether the same businesses and comparable cost measurements were kept.
>
> H4-A reports a difference between the two changes of about 812 AED per participant business: the participant businesses' change of -669 AED minus the comparison businesses' change of +143 AED. The model's 95% interval for the reduction is 792 to 833 AED. A programme-effect reading requires credible parallel trends and the other stated design conditions; one pre-period cannot show whether earlier trends were similar.
>
> Both the estimate and the interval are below our rule of at least a 1,000 AED reduction, if the model's assumptions hold. Precision does not resolve the causal assumption. I would request earlier measurements and evidence on competing changes before relying on the causal interpretation. This table alone does not justify claiming that the rule is met or that the result applies nationwide.

a. Match its numbers to H4-A (the difference-in-differences study summary) §2 and the table.

> Numbers: -669, +143 and about -812 (§2), interval 792 to 833 (table/§4). All appear in H4-A.

b. Match its conditions and requests for evidence to §3 and §4.

> Conditions (earlier measurements, competing changes, same businesses and comparable measures) match §3 and §4. Requests for earlier costs and competing changes are what §4 says the data cannot supply.

c. Does it separate the comparison, the credibility of the causal reading and the decision rule?

> Yes: the comparison (-812), the credibility of the causal reading (parallel trends, §3) and the decision rule (below 1,000) are three separate points.

d. Compare it with page 5. Which unsupported claim has been removed?

> Removed: "parallel trends is satisfied because both groups were followed over the same period", and "p-value and interval confirm ... GreenWaste caused the saving" and "clears the 1,000 AED minimum / wider rollout will meet this target".

e. Could any remaining wording be clearer?

> Open: e.g. say "mean" difference instead of "about 812 per business"; make clear that the interval is under the model assumptions. Accept any sensible edit.

Now return to page 4 and write your own interpretation and evidence request with the AI responses closed.